

TAHER MERCHANT

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Education

Delhi Technological University

B.Tech in Engineering Physics

July 2024 – July 2028

Experience

Lead Developer - Massachusetts Institute of Technology, CSAIL

June 2025 – Present

Guide: Prof. Manolis Kellis

- Pioneered a GitHub PR-to-Mantis integration that ingested and semantically mapped 60+ PRs and docs into cognitive cartography, enabling real-time tracking of ~40 active features and dependencies.
- Launched the Canvas Widget and Content Management System for Mantis, enabling collaborative content authoring, freehand annotation, semantic embedding mapping, and MantisGPT-powered agentic reasoning.

Research Intern - Carnegie Mellon University

June 2026 – Present

Guide: Prof. Min Xu

- Conducting research on mechanistic interpretability of MedSAM and SAM models, analyzing internal representations, feature activations, and decision-making mechanisms in deep learning and foundation models.
- Investigating interpretability methods including sparse autoencoders, activation patching, probing, and circuit analysis to understand model behavior, failure modes, and generalization across tasks and domains.

Founding Engineer - Avittam

February 2026 – Present

Stealth Startup | avittam.com

- Architected end-to-end system and database schema across 10+ Supabase tables, built and shipped the complete full-stack website with Next.js and REST APIs, hosted across GCP and Render serving 100+ users.
- Implemented 5+ automated Cron-based data pipelines cutting sync latency by 60%, and developed an inbuilt video meet platform supporting real-time sessions natively within the product.

Accomplishments

Google - GenAI Exchange Hackathon

November 2025

Top-10 National Rank

Leela Palace, Bangalore

- Designed an AI misinformation-detection system ranking Top 10 among 4,457+ submissions nationwide.

Best Project Implementation and Management Award - IRC'25

January 2025

Ranked 1st

BITS Pilani, Goa

- Won 1st place by leading development of an Arrow and Traffic Cone Detection Model using CNNs integrated with ROS for autonomous navigation.

Projects

TruthLens - Real-Time Multimodal Misinformation Verification | [LINK](#)

Nov 2025

- Spearheaded a real-time multimodal misinformation platform with dual inference pipelines, KG-based validation, and 3+ fact-checking APIs; Top 10 / 4,457+ teams at Google GenAI Hackathon.
- Productionized 6+ fine-tuned NLP and vision models as REST APIs backed by Supabase and Vercel, enabling real-time credibility scoring and analytics.

DRESS (Diffusion-based Recommender & Enhanced Styling System) | [LINK](#)

May 2025

- Designed a GAT recommender using BLIP-2 with MiniLM embeddings and triplet/contrastive losses, achieving high-precision retrieval on 11k+ fashion items with FAISS-based similarity search.
- Led development of a Diffusion-based Virtual Try-On/Off system with DensePose alignment and GAN-powered garment removal; achieved 90%+ SSIM realism.

Multi-Asset Regime-Aware Quantitative Trading System | [LINK](#)

Apr 2025

- Devised robust backtesting pipelines with SOTA feature engineering and statistical validation (Sharpe/Sortino ratios) using the Untrade SDK.
- Implemented a deep learning pipeline integrating 6+ technical indicators (RSI, MACD, Bollinger Bands) and 3 regime-switching algorithms, improving Sharpe Ratio by 0.7.

Publications

Zero Shot Trajectory Aware Retrieval on Semi Structured Knowledge Base via TARM-MCTS

2026

- *Under Review, AAAI 2027* | LLM-free MCTS retriever replacing LLM scorer with a cross-encoder compiled to OpenVINO IR, enabling deterministic CPU/NPU deployment at ~500ms latency with zero API calls at inference.

WeamRAG: Path Flexible Beam Search for Hierarchical Knowledge Graph Retrieval

2026

- *Under Review, EMNLP 2026* | Path-flexible hierarchical KG RAG framework combining Wu-Palmer-guided multi-seed beam search, binary-lifting LCA index, and BM25 chunk fallback; achieves Tok-F1 of 0.421 (leading all graph-structured baselines) and 94% pairwise LLM-judge win rate on MuSiQue (500 two-hop questions) at 0.95s retrieval over a 54,678-entity graph.

Technical Skills

Technologies: Python, C, Machine Learning, Deep Learning, Computer Vision, NLP, RaspberryPi 5, Google Cloud Platform

Frameworks & Libraries: NumPy, Pandas, Matplotlib, Seaborn, TensorFlow, Keras, OpenCV, HuggingFace, Linux

Certifications: [Deep Learning Specialization](#), [Machine Learning Specialization](#), [Intro to ML](#), [Intermediate ML](#)